Practical Lab 1: Setting Up Your Java Development Environment

Student-Centered, Self-Paced Guide

Lab Objectives

By the end of this lab, you will have:

1. Installed **VS Code** (your code editor)
2. Installed **Git** (your version control tool)
3. Created a **GitHub account** (your online code portfolio)
4. Connected VS Code to GitHub
5. Pushed your first Java program to GitHub

Time Required: 60–90 minutes
Difficulty: Beginner
Instructor Intervention: None – you will figure it out yourself!

Before You Begin

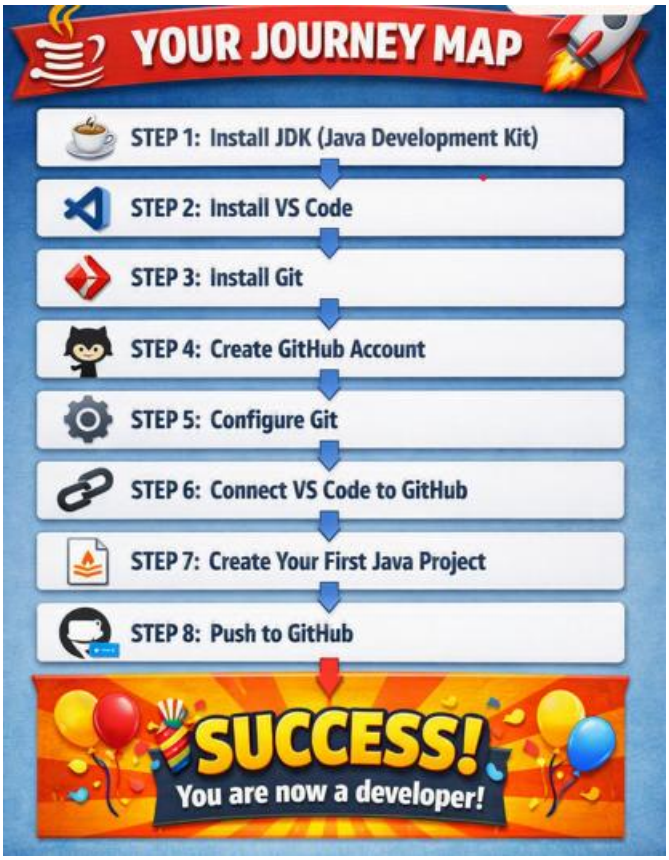
You will need:

- A computer running Windows, macOS, or Linux
- Internet connection
- An email address
- Patience and curiosity!

Golden Rule: If you get stuck, **Google the error message**. That's what real developers do. Only ask a classmate or instructor after you've tried:

1. Reading the error message carefully
2. Searching on Google
3. Watching a YouTube tutorial

Your Roadmap



STEP 1: Install JDK (Java Development Kit)

Why? Java code needs the JDK to compile and run.

Windows

1. Open your browser and go to: **https://adoptium.net/**
2. Click the big blue button that says **"Latest LTS Release"** (LTS = Long Term Support)
3. Download the **Windows x64 Installer** (.msi file)
4. Run the installer
5. **IMPORTANT:** Check the box that says **"Set JAVA_HOME variable"**
6. Click **Next → Next → Install**
7. Wait for installation to complete

🍎 macOS

Option A: Using Homebrew (Recommended)

1. Open **Terminal** (Press Cmd + Space, type "Terminal", press Enter)
2. Paste this command and press Enter:

bash

/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"

3. Wait for Homebrew to install (may take 10 minutes)
4. Then run:

bash

brew install openjdk@21

Option B: Manual Download

1. Go to: **https://adoptium.net/**
2. Download the **macOS Installer** (.pkg file)
3. Run the installer and follow the prompts

🐧 Linux (Ubuntu/Debian)

1. Open **Terminal** (Ctrl + Alt + T)
2. Run these commands one by one:

bash

sudo apt update

sudo apt install openjdk-21-jdk -y

🔧 Verify JDK Installation

Open your terminal/command prompt and type:

bash

java -version

Expected output (something like):

openjdk version "21.0.1" 2024-10-15

OpenJDK Runtime Environment (build 21.0.1+12)

OpenJDK 64-Bit Server VM (build 21.0.1+12, mixed mode)

- **If you see "command not found":**

- Windows: Restart your computer and try again
- macOS/Linux: Close and reopen Terminal

STEP 2: Install VS Code

Why? VS Code is your code editor – where you'll write Java programs.

All Operating Systems

1. Open your browser and go to: **https://code.visualstudio.com/**
2. Click the big **Download** button (it auto-detects your OS)
3. Run the installer:
    - **Windows:** Double-click the .exe file → Accept → Next → Next → Install
    - **macOS:** Open the .zip file → Drag **Visual Studio Code** to **Applications**
    - **Linux:** Install the .deb file using your software center, or run:

bash

sudo dpkg -i code_*.deb

- **Verify VS Code Installation**

1. Open VS Code
2. You should see the **Welcome** screen

- **Install Essential Extensions**

Inside VS Code:

1. Click the **Extensions** icon on the left sidebar (looks like 4 squares)
2. Search for and install these extensions (click **Install** for each):

- **Extension Pack for Java** (by Microsoft)
- **GitLens** (by GitKraken)
- **Prettier** (by Prettier)

Wait for each extension to finish installing before moving on.

STEP 3: Install Git

Why? Git tracks changes in your code and lets you push to GitHub.

Windows

1. Go to: https://git-scm.com/download/win
2. The download should start automatically
3. Run the .exe installer
4. **IMPORTANT:** Keep all default settings – just click **Next** repeatedly
5. Click **Install**

macOS

Option A: Using Homebrew

bash

brew install git

Option B: Using Xcode Command Line Tools

bash

xcode-select --install

A popup will appear – click **Install**.

Linux (Ubuntu/Debian)

bash

sudo apt install git -y

Verify Git Installation

Open terminal and type:

bash

git --version

Expected output:

text

git version 2.43.0

STEP 4: Create GitHub Account

Why? GitHub is your online portfolio – where employers will see your code.

1. Open your browser and go to: **https://github.com/**
2. Click **Sign up** (top right corner)
3. Enter:
    - **Email:** Use a professional email (e.g., firstname.lastname@gmail.com)
    - **Password:** Strong password (12+ characters)
    - **Username:** Use your real name (e.g., john-otieno or jotieno-dev)

✓ **IMPORTANT:** Choose a professional username – this is your developer identity!

4. Complete the puzzle/verification
5. Check your email and enter the **verification code**
6. On the "Welcome" screen, you can skip personalization
7. Choose the **Free** plan

✓ **Verify GitHub Account**

1. Go to: **https://github.com/YOUR-USERNAME**
    - Replace YOUR-USERNAME with your actual username
2. You should see your profile page

✓ **Personalize Your Profile (Optional but Recommended)**

1. Click the **Edit profile** button
2. Add:

- Your full name
- A short bio: "First-year Computer Science student learning Java"
- Your university

3. Click **Save**

STEP 5: Configure Git

Why? Git needs to know who you are so your commits are labeled correctly.

Open your terminal and run these commands one by one:

bash

git config --global user.name "Your Full Name"

bash

git config --global user.email "your.email@example.com"

Replace with your actual information. For example:

bash

git config --global user.name "John Otieno"

git config --global user.email "john.otieno@gmail.com"

- ✓ **Use the SAME email you used for GitHub!**
- ✓ **Verify Configuration**

bash

git config --list

You should see your name and email in the output.

STEP 6: Connect VS Code to GitHub

1. Open **VS Code**
2. Click the **Accounts** icon at the bottom-left (looks like a person)
3. Click **"Sign in with GitHub"**

4. A browser window opens – click **Authorize Visual Studio Code**
5. VS Code will show "Signed in as YOUR-USERNAME"

✓ **Verify Connection**

1. In VS Code, press Ctrl + Shift + P (Windows/Linux) or Cmd + Shift + P (Mac)
2. Type: GitHub: Sign in
3. You should see you're already signed in

STEP 7: Create Your First Java Project

7.1 Create a Folder for Your Course

1. Open **File Explorer** (Windows) or **Finder** (Mac) or **Files** (Linux)
2. Navigate to your **Documents** folder
3. Create a new folder called: java-course
4. Inside java-course, create a subfolder called: week1

7.2 Open the Folder in VS Code

1. Open **VS Code**
2. Click **File → Open Folder**
3. Select your java-course folder
4. Click **Select Folder** (Windows) or **Open** (Mac)

7.3 Create Your First Java File

1. In VS Code, right-click on the week1 folder in the sidebar
2. Click **New File**
3. Name it: HelloWorld.java
4. Press Enter

7.4 Write Your First Java Program

Copy and paste this code into HelloWorld.java:

java

```
/**
 * My first Java program
 * @author Your Name
 * @date Today's Date
 */
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
        System.out.println("My name is: [Your Name]");
        System.out.println("I am learning Java at [Your University]");
        System.out.println("Today's date is: [Today's Date]");
    }
}
```

Replace the bracketed text with your actual information.

7.5 Run Your Program

1. Press Ctrl + S to save the file
2. Click the **Run** button (▶️) at the top-right of VS Code
3. In the terminal at the bottom, you should see:

Output

```
Hello, World!
My name is: John Otieno
I am learning Java at University of Nairobi
Today's date is: 2026-01-15
```

- ✓ **Congratulations! You just ran your first Java program!**

STEP 8: Push to GitHub

8.1 Initialize Git in Your Project

1. In VS Code, click the **Source Control** icon on the left sidebar (looks like a branch)
2. Click **"Initialize Repository"**
3. You should see your files listed under "Changes"

8.2 Create a .gitignore File

Why? To prevent unnecessary files (like compiled .class files) from being uploaded.

1. Right-click on the java-course folder
2. Click **New File**
3. Name it: .gitignore
4. Paste this content:

text

```
# Compiled Java files
*.class


# VS Code settings
.vscode/


# OS files
.DS_Store
Thumbs.db


# Logs
*.log
```

5. Save the file (Ctrl + S)

8.3 Stage and Commit Your Changes

1. In VS Code, go to **Source Control** panel
2. You should see HelloWorld.java and .gitignore under "Changes"
3. Click the + icon next to each file to **stage** them
   (Or hover over "Changes" and click the + to stage all)
4. In the message box at the top, type: Initial commit: Hello World program
5. Click the ✓ **Commit** button

8.4 Create a GitHub Repository

1. Open your browser and go to **https://github.com/**
2. Click the + icon (top-right) → **New repository**
3. Fill in:
   - **Repository name:** java-course
   - **Description:** My Java learning journey - OOP Course
   - **Visibility:** ✅ **Public** (so employers can see your work)
   - **DO NOT** check "Add a README file"
4. Click **Create repository**

8.5 Connect Local Repository to GitHub

After creating the repository, GitHub will show you commands. In VS Code terminal (Ctrl + ~), run:

bash

```
git remote add origin https://github.com/YOUR-USERNAME/java-course.git
```

Replace YOUR-USERNAME **with your actual GitHub username.**

Then push:

bash

```
git branch -M main
```

```
git push -u origin main
```

8.6 Authenticate with GitHub

A browser window may open asking you to authorize Git. Click **Authorize**.

If prompted for credentials:

- ✓ **Username:** Your GitHub username
- ✓ **Password:** Use a **Personal Access Token** (not your GitHub password)
    - To create one: GitHub → Settings → Developer settings → Personal access tokens → Tokens (classic) → Generate new token
    - Select scope: repo (full control)
    - Copy the token and use it as your password
- ✓ **Verify Push Success**

1. Refresh your GitHub repository page
2. You should see:
    - HelloWorld.java
    - .gitignore
    - A commit message: "Initial commit: Hello World program"

    CONGRATULATIONS! You've pushed your first code to GitHub!

Final Self-Assessment Checklist

Tick each box only when you've verified it:

- □ JDK installed and java -version works
- □ VS Code installed with Java Extension Pack
- □ Git installed and git --version works
- □ GitHub account created and verified
- □ Git configured with name and email
- □ VS Code signed in to GitHub
- □ HelloWorld.java created and runs successfully

- □ .gitignore file created
- □ Local repository initialized
- □ First commit made
- □ GitHub repository created
- □ Code pushed to GitHub
- □ I can see my code on github.com

If you ticked all boxes → YOU ARE AMAZING!

➢ **Troubleshooting Guide**

Problem	Solution
java: command not found	Restart terminal/computer. Reinstall JDK and check "Set JAVA_HOME"
git: command not found	Restart terminal. Reinstall Git with default settings
VS Code won't run Java	Install "Extension Pack for Java" from Extensions marketplace
"Permission denied" on push	Use Personal Access Token instead of password
"Repository not found"	Check the URL: https://github.com/YOUR-USERNAME/java-course.git
Files not showing in Source Control	Save all files (Ctrl + S), then refresh VS Code
Merge conflict error	Don't panic! Run git pull origin main --rebase then git push
GitHub email not verified	Check spam folder. Resend verification email

- **Challenge Tasks (For Fast Finishers)**

If you finish early, try these:

1. **Create a second Java file** called AboutMe.java that prints 5 facts about you. Push it to GitHub.
2. **Modify** .gitignore to also ignore .pdf files.
3. **Add a README.md file** to your repository with:

markdown

My Java Course

About Me

- Name: [Your Name]

- University: [Your University]

- Course: Object-Oriented Programming

What I'm Learning

- Java fundamentals

- Object-oriented programming

- Git and GitHub

My Progress

- [x] Week 1: Setup complete

- [ ] Week 2: Variables and data types

4. **Create a folder** called week2 and push it (with a placeholder file).
5. **Explore GitHub**: Find 3 Java repositories you find interesting and star them.

- **Weekly Workflow (What You'll Do Every Week)**

From now on, every time you write code in class:



Commit Messages Should Be:

- Week 2: Variables and operators practice
- Week 3: Loop exercises - triangle patterns
- Week 4: OOP - BankAccount class

- **Reflection Questions**

After completing this lab, answer these in your notebook:

1. What was the most challenging part of this setup?
2. What error message did you encounter, and how did you solve it?
3. Why do developers use Git and GitHub?
4. How will this setup help you in your career?

➢ What You've Achieved

You have just set up the **exact same professional development environment** used by software engineers at:

- Safaricom
- Google
- Microsoft
- Andela
- Every major tech company in the world

You are no longer just a student - you are a developer.

➢ Next Steps

In your next class, you will:

1. Write Java programs using variables and data types
2. Push each program to your GitHub repository
3. Build a portfolio that will impress future employers

Remember: Every professional developer has a GitHub account. Yours is now live!

Additional Resources (Self-Help)

Topic	Resource
Git Basics	https://git-scm.com/book/en/v2/Getting-Started-Git-Basics
GitHub Guides	https://guides.github.com/

Topic	Resource
VS Code Java	https://code.visualstudio.com/docs/java/java-tutorial
Markdown Guide	https://www.markdownguide.org/basic-syntax/
Java Documentation	https://docs.oracle.com/en/java/

END OF PRACTICAL LAB 1

"The expert in anything was once a beginner." – Helen Hayes